Homework Assignment 9

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3. Suppose (X, S, μ) is a measure space. Suppose $h \in \mathcal{L}^1(\mu)$ and $||h||_1 > 0$. Prove that there is at most one number $c \in (0, \infty)$ such that

$$\mu\left(\{x \in X : |h(x)| \geq c\}\right) = \frac{1}{c}||h||_1.$$

Proof. For notational convenience, define $E_\alpha = \mu(\{x \in X : |h(x)| \geq \alpha\})$. Suppose that c and d both satisfy (3), and $c < d$. Then

$$\begin{aligned} \mu(\{x \in X : c \leq |h(x)| < d\}) &= E_c - E_d \\ &= \frac{1}{c}||h||_1 - \frac{1}{d}||h||_1 \\ &= \left(\frac{1}{c} - \frac{1}{d}\right)||h||_1 \end{aligned}$$

□

4. Show that the constant 3 in the Vitali Covering Lemma cannot be replaced by a smaller positive constant.

Proof. We will study the set of intervals $\{I_1 = (-1-\varepsilon, 1+\varepsilon), I_2 = (1, 3)\}$ for some choice of $\varepsilon > 0$. This collection is not already disjoint, so to satisfy the lemma, we would need to find one of them, I_α, such that cI_α contains the other. Multiplying these by a constant $c > 0$ yields

$$\begin{aligned} cI_1 &= (-c - c\varepsilon, c + c\varepsilon) \\ cI_2 &= (-c + 2, c + 2). \end{aligned}$$

Consider $0 < c < 1$. We want to see if we can find an $\varepsilon > 0$ such that $c + c\varepsilon \leq 1$ and $1 + \varepsilon \leq -c + 2$. This would make $I_1 \cap cI_2 = cI_1 \cap I_2 = \varnothing$. Indeed, just by manipulating the inequalities above, we get

$$\varepsilon \leq \frac{1-c}{c}$$

and

$$\varepsilon \leq 1 - c.$$

We choose the smaller of these, which is

$$\varepsilon = 1 - c$$

since $1/c > 1$. Since $1 - c > 0$, we have found a counterexample when $0 < c < 1$.

Now consider when $c \geq 1$. Then as we saw in the previous paragraph, since $\varepsilon > 0$, we always have the right end of cI_1 overlapping I_2, and likewise the left end of cI_2 overlaps I_1. Thus for the lemma to *not* be satisfied, we would need $c + 2 < 1$ and $\square$

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